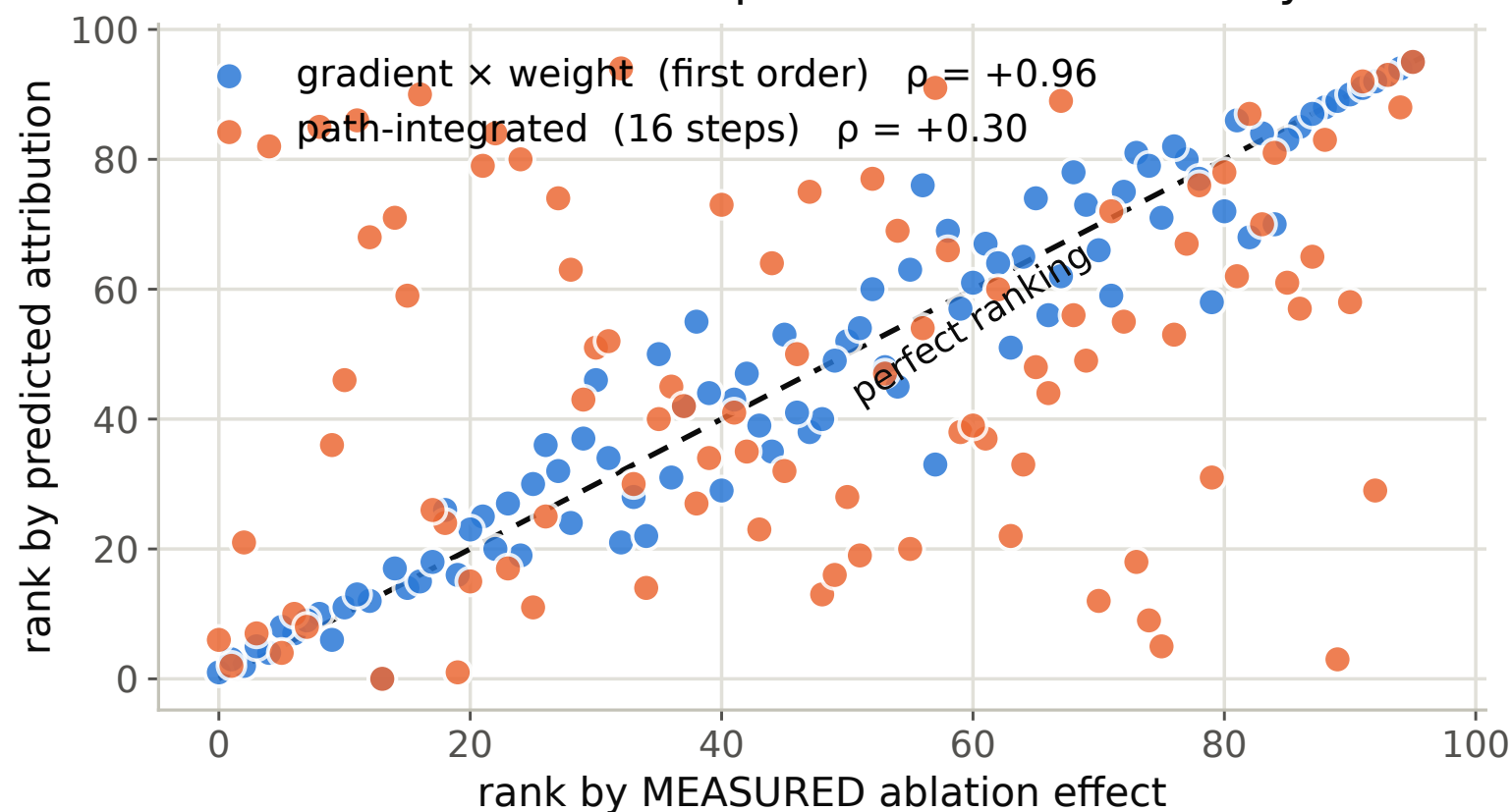


Is gradient attribution already causal necessity?

Largely yes — $dR/dW \cdot W$ summed over a component's owned entries ranks components by their true ablation effect at $\rho = 0.96$ ($n = 96$, sampled across the whole mass range; measured by actually deleting each one). So the information IS in the features. What removes it is the CLUSTERING: streaming_decomposition.py L2-normalises the feature vector before spherical k-means, and magnitude is exactly the part that encodes 'how much this matters'. What survives is direction — which weights fire together. The decomposition is co-attribution by construction.

Attribution does predict causal necessity



Integrating the path helps the SUM, hurts the RANKING

